PROBLEM 31

Multi-Particle Cable Constraint

# Two Particles on Cone with String

Original statement · Problems.pdf, p. 19

Two heavy point particles $P_1$ and $P_2$ of mass $m$ each are connected by an inextensible, massless string of total length $\ell$. Particle $P_1$ is constrained to move without friction over the surface of a cone of semiangle $\alpha$ (the particle cannot leave the surface of the cone). The string passes through an aperture at the cone vertex $O$, and it can slide through it without friction. Particle $P_2$ hangs along the cone axis, and is constrained to move along it. We define an inertial reference frame $S_0$. We call $\xi$ the length of the fraction of string between $O$ and $P_1$, and $\theta$ the angle formed between the $Ox$ axis and the projection onto the $Oxy$ plane of the $\mathbf{OP}_1$ vector. In the following, assume that the string is always tense and that the particles do not reach point $O$ during motion.

**(a)** Write down the general expression of the position, velocity, and acceleration vectors of $P_1$ and $P_2$ as a function of $\theta, \xi$ and the parameters of the problem.

**(b)** Write down the expression of all forces that act on $P_1$ and $P_2$.

**(c)** Write down the differential equations that, if solved, would yield the time evolution of $\theta, \xi$, the tension on the string, and any other reaction force in the problem.

**(d)** What is the mathematical condition that the function $\xi(t)$ must satisfy for the string to remain tense all the time?

**(e)** How fast should $P_1$ rotate in the $\theta$ direction to have a circular trajectory over a cone parallel? What is the value of the string tension and any other reaction force in the problem in that case? Give your answer as a function of $\xi$ and the parameters of the problem.

**(f)**

Taking these equations as the starting point (or based on the physical considerations of the problem) write down two first-integrals of the problem that do not depend on the string tension or any other reaction force in the problem.

**Full Analytical Solution (4-Phase Methodology)**

PHASE 1

### Physical Statement, Hypotheses & Parameters

- **Physical System:** Two heavy particles $P_1$ and $P_2$, each of mass $m$, connected by an inextensible massless string of length $\ell$.
- $P_1$ is constrained to the frictionless surface of a cone of semi-angle $\alpha$ with vertex at $O$.
- $P_2$ hangs vertically along the cone's axis of symmetry ($Oz$).
- The string passes through a frictionless hole at vertex $O$.
- **Configuration Degrees of Freedom (CDOFs):**
- Ambient coordinates: 6.
- Constraints: $P_1$ on cone (1), string inextensibility $\xi_1 + \xi_2 = \ell$ (1), $P_2$ on axis $Oz$ (2).
- Total CDOFs: $6 - 4 = 2$.
- Generalized coordinates: $\xi(t)$ (distance $OP_1$) and azimuthal angle $\theta(t)$.
- **Hypotheses:** String is always taut ($T(t) > 0$), frictionless hole and cone surface.
- **Parameters:** $m, \ell, \alpha, g > 0$.

PHASE 2

### Reference Frames, Vector Bases & Coordinate Geometry

- Cylindrical frame on the cone:
- Unit vector along cone generator from $O$ to $P_1$:

$$\mathbf{e}_\xi = \sin\alpha\,\mathbf{e}_r - \cos\alpha\,\mathbf{k}$$

- Unit tangent along parallel: $\mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$.

- Unit outward normal to the cone:

$$\mathbf{n} = \cos\alpha\,\mathbf{e}_r + \sin\alpha\,\mathbf{k}$$

- Position of $P_1$:

$$\mathbf{r}_1 = \xi\mathbf{e}_\xi = \xi\sin\alpha\,\mathbf{e}_r - \xi\cos\alpha\,\mathbf{k}$$

- Position of $P_2$ (hanging at distance $\ell - \xi$ below $O$):

$$\mathbf{r}_2 = -(\ell - \xi)\mathbf{k}$$

PHASE 3

### Step-by-Step Mathematical Deduction

**Part (a): Kinematics of $P_1$ and $P_2$**

1. **For $P_1$:**

Velocity:

$$\mathbf{v}_1 = \dot{\xi}\mathbf{e}_\xi + \xi\sin\alpha\dot{\theta}\mathbf{e}_\theta$$

Acceleration:

$$\mathbf{a}_1 = (\ddot{\xi} - \xi\dot{\theta}^2\sin^2\alpha)\mathbf{e}_\xi + (2\dot{\xi}\dot{\theta}\sin\alpha + \xi\ddot{\theta}\sin\alpha)\mathbf{e}_\theta - \xi\dot{\theta}^2\sin\alpha\cos\alpha\,\mathbf{n}$$

2. **For $P_2$:**

$$\mathbf{v}_2 = \dot{\xi}\mathbf{k}$$

$$\mathbf{a}_2 = \ddot{\xi}\mathbf{k}$$

**Part (b): Forces on $P_1$ and $P_2$**

1. On $P_1$:

- Gravity: $\mathbf{F}_{g1} = -mg\mathbf{k} = mg\cos\alpha\mathbf{e}_\xi - mg\sin\alpha\mathbf{n}$.
- String tension: $\mathbf{T}_1 = -T\mathbf{e}_\xi$ ($T > 0$).
- Normal cone reaction: $\mathbf{N} = N\mathbf{n}$.

2. On $P_2$:

- Gravity: $\mathbf{F}_{g2} = -mg\mathbf{k}$.
- String tension: $\mathbf{T}_2 = +T\mathbf{k}$.
- Guide reaction: $\mathbf{R}_2 \perp \mathbf{k}$.

**Part (c): Equations of Motion**

1. $P_2$ along $\mathbf{k}$:

$$m\ddot{\xi} = T - mg \implies T = m(g + \ddot{\xi})$$

2. $P_1$ along generator $\mathbf{e}_\xi$:

$$m(\ddot{\xi} - \xi\dot{\theta}^2\sin^2\alpha) = mg\cos\alpha - T$$

Substitute $T = m(g + \ddot{\xi})$:

$$m(\ddot{\xi} - \xi\dot{\theta}^2\sin^2\alpha) = mg\cos\alpha - m(g + \ddot{\xi})$$

$$2\ddot{\xi} - \xi\dot{\theta}^2 \sin^2\alpha + g(1-\cos\alpha) = 0$$

3. $P_1$ along parallel $\mathbf{e}_\theta$:

$$m(\xi\ddot{\theta}\sin\alpha + 2\dot{\xi}\dot{\theta}\sin\alpha) = 0 \implies \xi\ddot{\theta} + 2\dot{\xi}\dot{\theta} = 0$$

$$\frac{d}{dt}(\xi^2\dot{\theta}) = 0$$

4. $P_1$ along normal $\mathbf{n}$:

$$m\,\mathbf{a}_1 \cdot \mathbf{n} = N - mg\sin\alpha \implies N = mg\sin\alpha - m\xi\dot{\theta}^2\sin\alpha\cos\alpha$$

**Part (d): Condition for String to Remain Taut**

For the string to remain in tension ($T > 0$):

$$T = m(g + \ddot{\xi}) > 0 \implies \ddot{\xi} > -g$$

**Part (e): Circular Trajectory on a Parallel**

For a steady circular trajectory: $\xi = \text{constant} \implies \dot{\xi} = 0, \ddot{\xi} = 0$.

The radial equation gives:

$$-\xi\dot{\theta}^2\sin^2\alpha + g(1-\cos\alpha) = 0 \implies \xi\dot{\theta}^2\sin^2\alpha = g(1-\cos\alpha)$$

$$\dot{\theta} = \frac{1}{\sin\alpha}\sqrt{\frac{g(1-\cos\alpha)}{\xi}}$$

Values of Tension and Normal Force:

$$T = mg$$

$$N = mg\sin\alpha - m\cos\alpha\left[\frac{g(1-\cos\alpha)}{\sin\alpha}\right] = \frac{mg}{\sin\alpha}(\sin^2\alpha - \cos\alpha + \cos^2\alpha) = \frac{mg(1-\cos\alpha)}{\sin\alpha}$$

**Erratum:** The first version had $\mathbf{a}_1 \cdot \mathbf{n} = +\xi\dot{\theta}^2\sin\alpha\cos\alpha$ and $N = mg\sin\alpha + m\xi\dot{\theta}^2\sin\alpha\cos\alpha$ (its own algebra in (e) did not reach the quoted result). The centripetal term points against the outward normal: $\mathbf{a}_1 \cdot \mathbf{n} = -\xi\dot{\theta}^2\sin\alpha\cos\alpha$, hence $N = mg\sin\alpha - m\xi\dot{\theta}^2\sin\alpha\cos\alpha$, which gives the same final value $N = mg(1-\cos\alpha)/\sin\alpha$. Numeric check: sympy differentiation of $\mathbf{r}_1$ gives $\mathbf{a}_1 \cdot \mathbf{n} = -\tfrac{1}{2}\xi\dot{\theta}^2\sin 2\alpha$ and $N/(mg) = 0.5774$ at $\alpha = 60^\circ$, equal to $(1-\cos\alpha)/\sin\alpha$.

**Part (f): Two First Integrals of Motion**

1. **Angular Momentum First Integral:**

$$\xi^2\dot{\theta} = C_1$$

2. **Mechanical Energy First Integral:**

Kinetic energy of both particles:

$$T_{total} = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}m(\dot{\xi}^2 + \xi^2\dot{\theta}^2\sin^2\alpha) + \frac{1}{2}m\dot{\xi}^2 = m\dot{\xi}^2 + \frac{1}{2}m\xi^2\dot{\theta}^2\sin^2\alpha$$

Potential energy:

$$V_{total} = mgz_1 + mgz_2 = -mg\xi\cos\alpha - mg(\ell - \xi) = mg\xi(1 - \cos\alpha) - mg\ell$$

Conserved total energy $E = T_{total} + V_{total}$:

$$2\dot{\xi}^2 + \xi^2\dot{\theta}^2\sin^2\alpha + 2g\xi(1 - \cos\alpha) = C_2$$

Compare with the official solution key (Problems.pdf, Solution keys section).

PHASE 4

### Physical Interpretation & Dimensional Verification

- The effective potential $V_{eff}(\xi) = \frac{mC_1^2\sin^2\alpha}{2\xi^2} + mg\xi(1 - \cos\alpha)$ exhibits a stable minimum corresponding to the circular parallel orbit.

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*Source: Problems.pdf (Page 19)*